

Deep Learning for the Discovery of Small-Molecule Antimalarial Drug Leads

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Motivation & Questions

- Plasmodium falciparum* (Pf.) causes the deadliest form of Malaria and has the highest incidence of drug-resistance.
- While deep-learning (DL) methods dramatically accelerate early drug discovery, they also generating vast theoretical chemical spaces (10^{60} to 10^{80} molecules), most of which are synthetically infeasible or biologically inactive (Stokes et al., 2020).

Approach: Use of pIC₅₀ scores to train and validate direct message passing neural network (DMPNN) and analyse generalizability.

- Standardize ChEMBL *P. falciparum* IC₅₀ data to yield a high-quality dataset.
- Train DMPNN on asexual-stage compounds and test on sexual-stage data to analyse cross-stage predictive capability.

Questions: What is the preprocessing? How do data splits affect prediction outcomes within asexual-stage training data? Can models trained on asexual-stage data generalise to predict sexual-stage activity?

Background: Antimalarial drug discovery & DL methods

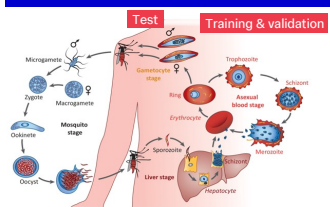


Figure 1. Malaria Life Cycle (Maier et al., 2019)

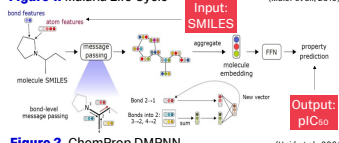
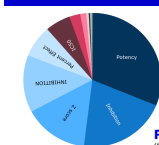


Figure 2. ChemProp DMPNN (Heid et al., 2024)

Dataset & Data Curation



- ChEMBL364 (snapshot 05/02/25)
- 969,219** molecules tested against *P. falciparum*.
- Generated from: PubChem, Gates Foundation, MalariaBox, and scientific literature.

Figure 3. ChEMBL364 Data (European Molecular Biology Laboratory - European Bioinformatics Institute, 2025)

Methods: Deep Learning Pipeline and Progression

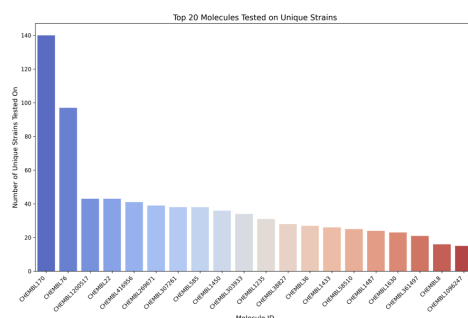


Methods: Dataset size after each preprocessing step



Results: Data after preprocessing

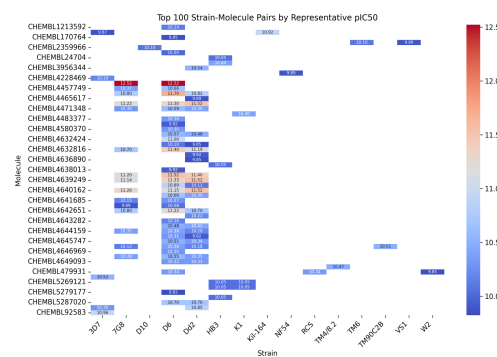
Data distribution



In the overall dataset:
19,230 unique molecules & **287** unique strains

Figure 4. Top 20 ChEMBL364 compounds ranked by number of unique *P. falciparum* strains they had been assayed on. ChEMBL170 (Quinine) and ChEMBL 76 (Chloroquine) exhibit the most strain coverage (140 and 97 strains respectively)

Molecular Potency: IC₅₀ → pIC₅₀



What is IC₅₀?
The concentration of a drug required to inhibit 50% of the biological activity for a target.
$$pIC_{50} = -\log_{10}\left(\frac{IC_{50}}{M}\right)$$

Figure 5. Heatmap of top 100 strain-molecule assays indicating both molecules with the strongest potency (pIC₅₀) are ChEMBL4457749 tested on strains 7G8 and D6. White spaces are strain-molecule pairs not found in dataset.

Results cont'd

Chemical Space of Processed ChEMBL Data

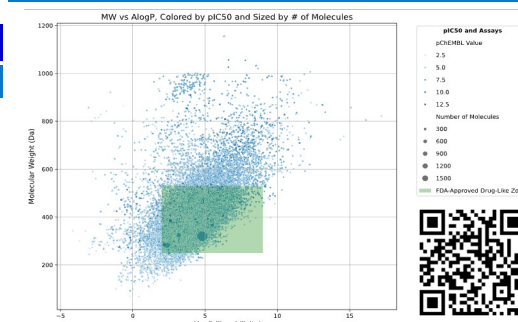


Figure 6. Physicochemical profile of ChEMBL364 (n= 39,907). Most compounds cluster around with molecular weights within 250-500 Da and AlogP of 0-5. A handful of highly lipophilic and/or exceptionally large molecules show variable potency.

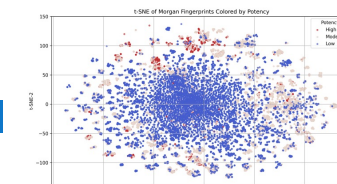


Figure 7. t-SNE embedding of Morgan fingerprints illustrating antimalarial compound clustering patterns based on structural similarity with notable segregation of high-potency antimalarials.

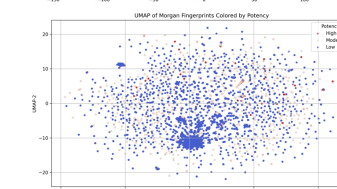


Figure 8. UMAP dimensionality reduction of Morgan fingerprint descriptors reveals heterogeneous distribution of antimalarial activity across the chemical space.

Future Directions

- Complete the train, validate, test, and molecule prediction steps.
- Benchmark data splits (scaffold, per strain, by malarial stage).
- Optimise hyperparameters
- Assess descriptor and atom-feature importance by removing/masking singular input features.